

- HDD video: [What's inside of a hard drive?](#)
- Indexed Allocation: [OS69 - Indexed Allocation | File Storage](#)
- [Turn on screen reader support](#)
- Banner hidden
- To enable screen reader support, press Ctrl+Alt+Z To learn about keyboard shortcuts, press Ctrl+slash
- What's inside of a hard drive?
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- Unix inode problems:
 - [Unix Inode file system](#)
 - In a filesystem where an inode has 12 direct block pointers, 1 single indirect pointer, and 1 double indirect pointer, and each block is 512 bytes and disk block addresses are 8 bits, the maximum file size ? **ANS: 128MB**
 - A file system uses UNIX inode data structure which contains 12 direct block addresses, 3 single indirect blocks and 2 triple indirect blocks. The size of each block is 64 Bytes and the size of each block address is 8 Bytes. Find the number of double indirect blocks if the maximum possible file size is 82.25 KB. **ANS: 4**
- **VSFS video:** [Operating Systems Lecture 19: File system implementation](#)
- **VSFS Math:**

A file system has inode size = 512B and block size = 4KB. First 3 blocks contain superblock, data bitmap and inode bitmap. Now calculate the address of the inode number 23. The disk is sector addressable and the size of each sector is 256 B. Find out the sector address of the inode.

Solution:

Inode start address = $3 \times 4 \text{ KB} = 12 \text{ KB}$

Offset = $23 \times 512 \text{ B} = 11.5 \text{ KB}$

Address of inode 23 = $12 + 11.5 \text{ KB} = 23.5 \text{ KB} \approx 23 \text{ KB}$

Block number = $23 \times 512 \text{ B} / 4 \text{ KB} = 11.5 \text{ KB} / 4 \text{ KB} = 2.875 \approx 2$

Sector address = $\{(2 \times 4 \text{ KB}) + 12 \text{ KB}\} / 256 \text{ B} = 80$

- **How Read Write occurs in file system:**
 - [How Does Your Computer Save and Read Files?](#)

- **Write Cost Comparison:**

I/O Cost Table			
Operation	Step	What Happens	# I/Os
1. Overwrite existing block	Read inode block	Needed to locate existing data block	1
	Write data block	Write new contents to existing block	1
Total			2 I/Os
2. Append (allocate new block)	Read block bitmap	Find free block	1
	Write block bitmap	Mark block as used	1
	Read inode block	Needed to add new block pointer	1
	Write inode block	Update block pointer, size, timestamps	1
	Write data block	Write actual data	1
Total			5 I/Os

- **Journaling:**

- Slide:  **Journaling.pdf**

3. Create new file	Read inode bitmap	Find free inode	1
	Write inode bitmap	Mark inode as used	1
	Read block bitmap	Find free block for file data	1
	Write block bitmap	Mark block as used	1
	Read parent directory block	Find space for new entry	1
	Write parent directory block	Add entry (filename, inode #)	1
	Write inode block	Initialize inode metadata	1
	Write data block	Write file contents	1
Total			8 I/Os